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HIGH FREQUENCY AMPLIFIER

Abstract

A high frequency amplifier includes a driver amplifier that amplifies an input high frequency signal and an asymmetrical Doherty amplifier that amplifies a high frequency signal output from the driver amplifier. The high frequency amplifier includes a first layered structure, a second layered structure layered on the first layered structure, and a third layered structure layered on the second layered structure. The asymmetrical Doherty amplifier includes a carrier amplifier, a peak amplifier, a branch circuit, a first signal line, a second signal line, and a phase adjusting circuit. The second layered structure includes a first wiring layer, a second wiring layer, a dielectric layer, and one or more capacitors. The driver amplifier, the carrier amplifier, and the peak amplifier are provided in the first layered structure. The branch circuit and the phase adjusting circuit are provided in the third layered structure.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority based on Japanese Patent Application No. 2024-035307 filed on Mar. 7, 2024, and the entire contents of the Japanese patent application are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a high frequency amplifier.

BACKGROUND

[0003] In recent years, broader bandwidths of mobile communication systems, such as mobile phones, are developed. Thus, for a power amplifier used in base station equipment of the system and the like, it is desirable to improve the power efficiency in a high frequency band and the like. As power amplifiers for achieving the improvement of the power efficiency, an asymmetrical Doherty amplifier including a carrier amplifier and a peak amplifier is known. The carrier amplifier may be referred to as a main amplifier. The asymmetrical Doherty amplifier is connected at a stage following a driver amplifier. Further, a high frequency amplifier in which a driver amplifier, a carrier amplifier, and a peak amplifier are three-dimensionally mounted has been proposed. [0004] Patent literature 1: Japanese Unexamined Patent Application Publication No. 2021-170703

SUMMARY

[0005] A high frequency amplifier of the present disclosure includes a driver amplifier configured to amplify an input high frequency signal and an asymmetrical Doherty amplifier configured to amplify a high frequency signal output from the driver amplifier. The high frequency amplifier includes a first layered structure, a second layered structure layered on the first layered structure, and a third layered structure layered on the second layered structure. The asymmetrical Doherty amplifier includes: a carrier amplifier; a peak amplifier configured to initiate an amplifying operation in a case where an output of the carrier amplifier has reached a saturation region, the peak amplifier having a saturation output different from a saturation output of the carrier amplifier; a branch circuit configured to receive the high frequency signal output from the driver amplifier; a first signal line provided between the branch circuit and the carrier amplifier; a second signal line provided between the branch circuit and the peak amplifier; and a phase adjusting circuit provided in either one or both of the first signal line and the second signal line and configured to delay either one or both of a phase of an input signal to the carrier amplifier and a phase of an input signal to the peak amplifier. The branch circuit is configured to branch the high frequency signal output from the driver amplifier into the first signal line and the second signal line. The second layered structure includes: a first wiring layer provided between the first layered structure and the third layered structure; a second wiring layer provided between the first wiring layer and the third layered structure; a dielectric layer provided between the first wiring layer and the second wiring layer; and one or more capacitors including a part of the first wiring layer, a part of the second wiring layer, and a part of the dielectric layer. The driver amplifier, the carrier amplifier, and the peak amplifier are provided in the first layered structure. The branch circuit and the phase adjusting circuit are provided in the third layered structure. The one or more capacitors are each electrically connected to any one of the driver amplifier, the carrier amplifier, and the peak amplifier.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a block diagram illustrating a high frequency amplifier according to an embodiment.

[0007] FIG. 2 is a circuit diagram illustrating a configuration of a driver amplifier, an input matching circuit, an output matching circuit, and an output bias circuit.

[0008] FIG. 3 is a circuit diagram illustrating a configuration of an asymmetrical Doherty amplifier.

[0009] FIG. 4 is a cross-sectional view of a high frequency amplifier according to the embodiment.

[0010] FIG. 5 is a first diagram illustrating a layout of a wiring layer.

[0011] FIG. 6 is a second diagram illustrating a layout of a wiring layer.

[0012] FIG. 7 is a third diagram illustrating a layout of a wiring layer.

[0013] FIG. 8 is a fourth diagram illustrating a layout of a wiring layer.

[0014] FIG. 9 is a fifth diagram illustrating a layout of a wiring layer.

[0015] FIG. 10 is a sixth diagram illustrating a layout of a wiring layer.

[0016] FIG. 11 is a first cross-sectional view illustrating a method of manufacturing a high frequency amplifier according to the embodiment.

[0017] FIG. 12 is a second cross-sectional view illustrating a method of manufacturing a high frequency amplifier according to the embodiment.

[0018] FIG. 13 is a third cross-sectional view illustrating a method of manufacturing a high frequency amplifier according to the embodiment.

[0019] FIG. 14 is a fourth cross-sectional view illustrating a method of manufacturing a high frequency amplifier according to the embodiment.

[0020] FIG. 15 is a fifth cross-sectional view illustrating a method of manufacturing a high frequency amplifier according to the embodiment.

[0021] FIG. 16 is a sixth cross-sectional view illustrating a method of manufacturing a high frequency amplifier according to the embodiment.

[0022] FIG. 17 is a seventh cross-sectional view illustrating a method of manufacturing a high frequency amplifier according to the embodiment.

[0023] FIG. 18 is a diagram illustrating a layout in a first example.

[0024] FIG. 19 is a diagram illustrating a layout in a second example.

DETAILED DESCRIPTION

[0025] In recent years, there has been an increasing demand for improvement in high frequency characteristics of a high frequency amplifier.

[0026] According to the present disclosure, high frequency characteristics can be improved.

Description of Embodiments of Present Disclosure

[0027] First, embodiments of the present disclosure will be listed and described.

[0028] (1) A high frequency amplifier according to an aspect of the present disclosure includes a driver amplifier configured to amplify an input high frequency signal and an asymmetrical Doherty amplifier configured to amplify a high frequency signal output from the driver amplifier. The high frequency amplifier includes a first layered structure, a second layered structure layered on the first layered structure, and a third layered structure layered on the second layered structure. The asymmetrical Doherty amplifier includes: a carrier amplifier; a peak amplifier configured to initiate an amplifying operation in a case where an output of the carrier amplifier has reached a saturation region, the peak amplifier having a saturation output different from a saturation output of the carrier amplifier; a branch circuit configured to receive the high frequency signal output from the driver amplifier; a first signal line provided between the branch circuit and the carrier amplifier; a second signal line provided between the branch circuit and the peak amplifier; and a phase adjusting circuit provided in either one or both of the first signal line and the second signal line and configured to delay either one or both of a phase of an input signal to the carrier amplifier and a phase of an input signal to the peak amplifier. The branch circuit is configured to branch the high frequency signal output from the driver amplifier into the first signal line and the second signal line. The second layered structure includes: a first wiring layer provided between the first layered structure and the third layered structure; a second wiring layer provided between the first wiring

layer and the third layered structure; a dielectric layer provided between the first wiring layer and the second wiring layer; and one or more capacitors including a part of the first wiring layer, a part of the second wiring layer, and a part of the dielectric layer. The driver amplifier, the carrier amplifier, and the peak amplifier are provided in the first layered structure. The branch circuit and the phase adjusting circuit are provided in the third layered structure. The one or more capacitors are each electrically connected to any one of the driver amplifier, the carrier amplifier, and the peak amplifier.

[0029] A capacitor including a part of the first wiring layer, a part of the second wiring layer, and a part of the dielectric layer is provided between the driver amplifier, the carrier amplifier, and the peak amplifier, and the branch circuit and the phase adjusting circuit. Thus, signal interference is less likely to occur between the driver amplifier, the carrier amplifier, and the peak amplifier, and the branch circuit and the phase adjusting circuit, and the high frequency characteristics can be improved.

[0030] (2) In (1), the high frequency amplifier may further include a first input matching circuit including the one or more capacitors and connected to an input terminal of the driver amplifier. In this case, the number of surface mount components included in the first input matching circuit can be reduced.

[0031] (3) In (2), the first input matching circuit may be provided in the first layered structure and the second layered structure. In this case, a part of the first wiring layer may be used as an electrode of the capacitor included in the first input matching circuit.

[0032] (4) In any one of (1) to (3), the high frequency amplifier may further include a second input matching circuit including the one or more capacitors and connected to an input terminal of the carrier amplifier. In this case, the number of surface mount components included in the second input matching circuit can be reduced.

[0033] (5) In (4), the second input matching circuit may be provided in the first layered structure and the second layered structure. In this case, a part of the first wiring layer may be used as an electrode of the capacitor included in the second input matching circuit.

[0034] (6) In any one of (1) to (5), the high frequency amplifier may further include a third input matching circuit including the one or more capacitors and connected to an input terminal of the peak amplifier. In this case, the number of surface mount components included in the third input matching circuit can be reduced.

[0035] (7) In (6), the third input matching circuit may be provided in the first layered structure and the second layered structure. In this case, a part of the first wiring layer may be used as an electrode of the capacitor included in the third input matching circuit.

[0036] (8) In any one of (1) to (7), the first wiring layer may have a ground region overlapping, in a plan view, at least with a transmission line through which a high frequency signal output from the carrier amplifier or the peak amplifier is transmitted. In particular, the high frequency signal output from the carrier amplifier or the peak amplifier is likely to be affected by the parasitic capacitance, but the ground region is provided, and thus it is easy to reduce deterioration of the high frequency signal due to the parasitic capacitance.

[0037] (9) In any one of (1) to (8), when n is an integer of 0 or more, an electrical length from an output terminal of the driver amplifier to an input terminal of the carrier amplifier may be within a range of $(2n+1)\times\pi-\pi/2$ to $(2n+1)\times\pi+\pi/2$ in terms of a phase of the high frequency signal. In this case, even when the driver amplifier and the carrier amplifier are close to each other, the electrical characteristics of the carrier amplifier can be easily stabilized.

[0038] (10) In any one of (1) to (9), when n is an integer of 0 or more, an electrical length from an output terminal of the driver amplifier to an input terminal of the peak amplifier may be within a range of $(2n+1)\times\pi-\pi/2$ to $(2n+1)\times\pi+\pi/2$ in terms of a phase of the high frequency signal. In this case, even when the driver amplifier and the peak amplifier are close to each other, the electrical characteristics of the carrier amplifier can be easily stabilized.

[0039] (11) In any one of (1) to (10), the saturation output of the peak amplifier may be larger than the saturation output of the carrier amplifier. In the peak amplifier, the phase shift amount for obtaining the optimum matching is larger than that in the carrier amplifier.

Details of Embodiments of Present Disclosure

[0040] Hereinafter, embodiments of the present disclosure will be described in detail, but the present disclosure is not limited thereto. In the specification and the drawings, components having substantially the same functional configuration are denoted by the same reference numerals, and redundant description thereof may be omitted. In the following description, an XYZ orthogonal coordinate system is used. However, the coordinate system is defined for description and does not limit the posture of the semiconductor device. Further, when viewed from an arbitrary point, a +Z side may be referred to as an upward side, an upper side, or a top, and a -Z side may be referred to as a downward side, a lower side, or a bottom.

(Circuit of High Frequency Amplifier)

[0041] First, a circuit of a high frequency amplifier according to an embodiment will be described. FIG. 1 is a block diagram illustrating a high frequency amplifier according to the embodiment.

[0042] A high frequency amplifier 1 according to the embodiment is mounted in a communication device, such as base station equipment of a mobile communication system. High frequency amplifier 1 is used to amplify a transmission signal, for example. High frequency amplifier 1 amplifies, for example, radio frequency (RF) signals having a frequency of about 5 GHz to 6 GHz.

[0043] Focusing on the circuit configuration, as illustrated in FIG. 1, high frequency amplifier 1 includes an input terminal RFin, an output terminal RFout, a driver amplifier 10, and an asymmetrical Doherty amplifier 40. Asymmetrical Doherty amplifier 40 includes a branch circuit 41, a first signal line 41A, a second signal line 41B, a carrier amplifier 20, and a peak amplifier 30. High frequency amplifier 1 further includes an input matching circuit 11, an output matching circuit 12, and an output bias circuit 14. Asymmetrical Doherty amplifier 40 further includes a phase adjusting circuit 23, an input matching circuit 21, an output matching circuit 22, an input bias circuit 24, a phase adjusting circuit 33, an input matching circuit 31, an output matching circuit 32, an input bias circuit 34, and a Doherty network 42.

[0044] Input matching circuit 11 is connected to input terminal RFin. Input matching circuit 11 also functions as an input bias circuit. Input matching circuit 11 is connected to an input terminal 18 of driver amplifier 10, and output matching circuit 12 and output bias circuit 14 are connected to an output terminal 19 of driver amplifier 10. An RF signal is input to driver amplifier 10 from input terminal RFin via input matching circuit 11, and driver amplifier 10 amplifies the input RF signal. Driver amplifier 10 amplifies the RF signal to such an extent that asymmetrical Doherty amplifier 40 can amplify the RF signal to a predetermined transmission power, for example. The RF signal amplified by driver amplifier 10 is output via output matching circuit 12. Input matching circuit 11 is an example of a first input matching circuit.

[0045] The RF signal output via output matching circuit 12 is input to branch circuit 41. First signal line 41A is provided between branch circuit 41 and carrier amplifier 20, and second signal line 41B is provided between branch circuit 41 and peak amplifier 30. Branch circuit 41 branches the input RF signal into first signal line 41A and second signal line 41B. For example, branch circuit 41 is a Wilkinson type divider, and equally branches the RF signal amplified by driver amplifier 10 into first signal line 41A and second signal line 41B.

[0046] Phase adjusting circuit 23 and input matching circuit 21 are provided in first signal line 41A. Input matching circuit 21 and input bias circuit 24 are connected to an input terminal 28 of carrier amplifier 20, and output matching circuit 22 is connected to an output terminal 29 of carrier amplifier 20. Phase adjusting circuit 23 delays the phase of a signal (an input signal to carrier amplifier 20) output from branch circuit 41 to first signal line 41A. The RF signal is input to carrier amplifier 20 via input matching circuit 21, and carrier amplifier 20 amplifies the input RF signal. The RF signal amplified by carrier amplifier 20 is output via output matching circuit 22. Input

matching circuit **21** is an example of a second input matching circuit.

[0047] Phase adjusting circuit **33** and input matching circuit **31** are provided in second signal line **41B**. Input matching circuit **31** and input bias circuit **34** are connected to an input terminal **38** of peak amplifier **30**, and output matching circuit **32** is connected to an output terminal **39** of peak amplifier **30**. Phase adjusting circuit **33** delays the phase of a signal (an input signal to peak amplifier **30**) output from branch circuit **41** to second signal line **41B**. The RF signal is input to peak amplifier **30** via input matching circuit **31** and peak amplifier **30** amplifies the input RF signal. Peak amplifier **30** initiates an amplifying operation in a case where an output of carrier amplifier **20** has reached a saturation region, and has a saturation output different from a saturation output of carrier amplifier **20**. The RF signal amplified by peak amplifier **30** is output via output matching circuit **32**. Input matching circuit **31** is an example of a third input matching circuit.

[0048] The RF signal output via output matching circuit **22** and the RF signal output via output matching circuit **32** are input to Doherty network **42**.

[0049] Next, a circuit configuration of driver amplifier **10**, input matching circuit **11**, output matching circuit **12**, and output bias circuit **14** will be described. FIG. **2** is a circuit diagram illustrating a configuration of driver amplifier **10**, input matching circuit **11**, output matching circuit **12**, and output bias circuit **14**.

[0050] Input matching circuit **11** includes capacitors **C2**, **C3**, and **C4** and inductors **L1** and **L2**. Capacitor **C3** and inductor **L1** are connected in series to input terminal **RFin** in this order. A gate of driver amplifier **10** is connected to a stage following inductor **L1**. Inductor **L2** is connected in parallel to capacitor **C3**. Capacitor **C2** is connected between the ground and a node between capacitor **C3** and inductor **L1**. Capacitor **C4** is connected between the ground and a node between inductor **L1** and the gate of driver amplifier **10**.

[0051] Output matching circuit **12** includes a capacitor **C8** and inductors **L4**, **L5**, **L9** and **L10**. Capacitor **C8**, inductor **L10**, and inductor **L9** are connected in series to the drain of driver amplifier **10** in this order. Inductor **L4** is connected between the ground and a node between inductor **L10** and inductor **L9**. Inductor **L5** is connected between inductor **L9** and the ground in a stage following inductor **L9**.

[0052] Output bias circuit **14** includes a capacitor **C6** and an inductor **L3**. Inductor **L3** is connected between a node to which a voltage **Vd** is input and the drain of driver amplifier **10**. Capacitor **C6** is connected between a node to which voltage **Vd** is input and the ground. Voltage **Vd** is supplied as a drain bias of driver amplifier **10** via inductor **L3**. Capacitor **C6** is a bypass capacitor for voltage **Vd**.

[0053] Next, asymmetrical Doherty amplifier **40** will be described. Asymmetrical Doherty amplifier **40** further amplifies the RF signal amplified by driver amplifier **10** and outputs the amplified RF signal from output terminal **RFout**. FIG. **3** is a circuit diagram illustrating a configuration of asymmetrical Doherty amplifier **40**.

[0054] As described above, asymmetrical Doherty amplifier **40** includes branch circuit **41**, first signal line **41A**, second signal line **41B**, carrier amplifier **20**, peak amplifier **30**, phase adjusting circuit **23**, input matching circuit **21**, output matching circuit **22**, input bias circuit **24**, phase adjusting circuit **33**, input matching circuit **31**, output matching circuit **32**, input bias circuit **34**, and Doherty network **42**.

[0055] Branch circuit **41** includes capacitors **C23**, **C24**, and **C29**, inductors **L11** and **L12**, and a resistor element **R3**. Inductor **L11** is connected between output matching circuit **12** and phase adjusting circuit **23**. Inductor **L12** is connected between output matching circuit **12** and phase adjusting circuit **33**. Capacitor **C23** is connected between the ground and a node between output matching circuit **12** and inductors **L11** and **L12**. Capacitor **C24** is connected between inductor **L11** and the ground at a stage following inductor **L11**. Capacitor **C29** is connected between inductor **L12** and the ground at a stage following inductor **L12**. Resistor element **R3** is connected between inductor **L11** and inductor **L12** at a stage following capacitors **C24** and **C29**.

[0056] Phase adjusting circuit **23** includes a capacitor **C30** and inductors **L13** and **L14**. Inductors

L13 and **L14** are connected in series to inductor **L11** in this order. Capacitor **C30** is connected between the ground and a node between inductor **L13** and inductor **L14**.

[0057] Phase adjusting circuit **33** includes capacitors **C32** and **C1** and an inductor **L15**. Capacitors **C32** and **C1** are connected in series to inductor **L12** in this order. Inductor **L15** is connected between the ground and a node between capacitor **C32** and capacitor **C1**.

[0058] Input matching circuit **21** includes capacitors **C11**, **C12**, and **C7** and an inductor **L17**. Capacitor **C11** and inductor **L17** are connected in series to inductor **L14** in this order. A gate of carrier amplifier **20** is connected to a stage following inductor **L17**. Capacitor **C12** is connected between the ground and a node between capacitor **C11** and inductor **L17**. Capacitor **C7** is connected between the ground and a node between inductor **L17** and the gate of carrier amplifier **20**.

[0059] Input bias circuit **24** includes a capacitor **C15**, a resistor element **R4**, and an inductor **L6**. Resistor element **R4** and inductor **L6** are connected in series in this order between a power source **Vg** and the gate of carrier amplifier **20**. Capacitor **C15** is connected between the ground and a node between power source **Vg** and resistor element **R4**. A gate bias of carrier amplifier **20** is supplied from power source **Vg** via inductor **L6**. Capacitor **C15** is a bypass capacitor of power source **Vg**, and resistor element **R4** is a resistor for adjustment.

[0060] Input matching circuit **31** includes capacitors **C10**, **C5**, and **C13** and an inductor **L7**. Capacitor **C10** and inductor **L7** are connected in series to capacitor **C1** in this order. A gate of peak amplifier **30** is connected to a stage following inductor **L7**. Capacitor **C5** is connected between the ground and a node between capacitor **C10** and inductor **L7**. Capacitor **C13** is connected between the ground and a node between inductor **L7** and the gate of peak amplifier **30**.

[0061] Input bias circuit **34** includes a capacitor **C21**, a resistor element **R5**, and an inductor **L8**. Resistor element **R5** and inductor **L8** are connected in series in this order between power source **Vg** and the gate of peak amplifier **30**. Capacitor **C21** is connected between the ground and a node between power source **Vg** and resistor element **R5**.

[0062] Output matching circuit **22** includes capacitors **C9** and **C26**. Capacitor **C26** is connected between the drain of carrier amplifier **20** and Doherty network **42**. Capacitor **C26** is a capacitor for blocking a DC component. Capacitor **C9** is connected between the ground and a node between capacitor **C26** and Doherty network **42**. Voltage **Vd** is supplied from an external output bias circuit as a drain bias of carrier amplifier **20**.

[0063] Output matching circuit **32** includes a capacitors **C28** and **C16** and an inductor **L16**. Capacitor **C28** and inductor **L16** are connected in series in this order between the drain of peak amplifier **30** and Doherty network **42**. Capacitor **C28** is a capacitor for blocking a DC component. Capacitor **C16** is connected between the ground and a node between the drain of peak amplifier **30** and capacitor **C28**. Voltage **Vd** is supplied from an external output bias circuit as a drain bias of peak amplifier **30**.

[0064] Doherty network **42** includes a capacitor **C25**. Capacitor **C25** is connected between the ground and a node between output terminal **RFout**, capacitor **C26** and inductor **L16**.

(Structure of High Frequency Amplifier)

[0065] Next, the structure of high frequency amplifier **1** according to the embodiment will be described. FIG. **4** is a cross-sectional view of a high frequency amplifier according to the embodiment. Note that FIG. **4** illustrates a schematic cross section of high frequency amplifier **1**, and the arrangement of driver amplifier **10**, carrier amplifier **20**, peak amplifier **30**, the surface mount component, the transmission line pattern, and the conductive via does not correspond to the circuits illustrated in FIGS. **1** to **3**.

[0066] Focusing on the cross-sectional configuration, as illustrated in FIG. **4**, high frequency amplifier **1** includes a first layered structure **111**, a second layered structure **112**, and a third layered structure **113**. Second layered structure **112** is provided on or over first layered structure **111**, and third layered structure **113** is provided on or over second layered structure **112**. First layered

structure **111**, second layered structure **112**, and third layered structure **113** have, for example, a square planar shape with each side having a length of, for example, 6 mm.

[0067] First layered structure **111** includes a wiring layer **121**, a dielectric layer **131**, a wiring layer **122**, a dielectric layer **132**, driver amplifier **10**, carrier amplifier **20**, peak amplifier **30**, and a plurality of surface mount components **51**.

[0068] Wiring layer **122** is provided on the lower surface of dielectric layer **132**. A thickness of dielectric layer **132** is, for example, 0.1 mm to 0.2 mm. For example, dielectric layer **132** has a relative dielectric constant of 3.0 to 3.7 and a dielectric loss tangent of 0.002 to 0.007. A thickness of wiring layer **122** is, for example, 10 μm to 45 μm . The thickness of wiring layer **122** may be 35 μm to 45 μm . Wiring layer **122** is, for example, a copper layer. As will be described in detail later, a transmission line pattern and a ground pattern are formed in wiring layer **122**. The conductivity of copper is about 6.25×10^{17} S/m.

[0069] Driver amplifier **10** is mounted on the lower surface of wiring layer **122**. Driver amplifier **10** is, for example, an amplifier including a gallium nitride (GaN)-based high electron mobility transistor (HEMT). Driver amplifier **10** includes a substrate and a semiconductor layer provided on or over the substrate. Driver amplifier **10** has a three dimensional shape of a rectangular parallelepiped shape including a main surface **10a** and a main surface **10b** opposite to main surface **10a**. Main surface **10a** is in the semiconductor layer, and main surface **10b** is in the substrate. An amplifier circuit including a semiconductor layer is formed on main surface **10a**. In main surface **10a**, a gate pad is provided in the vicinity of one of two sides parallel to each other, and a drain pad is provided in the vicinity of the other side. Further, source pads are provided on both sides of the gate pad, and the source pads are connected to electrodes provided on main surface **10b**. Main surface **10a** faces upward, and main surface **10b** faces downward.

[0070] Carrier amplifier **20** is mounted on the lower surface of wiring layer **122**. Carrier amplifier **20** is, for example, an amplifier including a GaN-based HEMT. Carrier amplifier **20** includes a substrate and a semiconductor layer provided on or over the substrate. Carrier amplifier **20** has a three dimensional shape of a rectangular parallelepiped shape including a main surface **20a** and a main surface **20b** opposite to main surface **20a**. Main surface **20a** is in the semiconductor layer, and main surface **20b** is in the substrate. An amplifier circuit including a semiconductor layer is formed on main surface **20a**. In main surface **20a**, a gate pad is provided in the vicinity of one of two sides parallel to each other, and a drain pad is provided in the vicinity of the other side. Further, source pads are provided on both sides of the gate pad, and the source pads are connected to electrodes provided on main surface **20b**. Main surface **20a** faces upward, and main surface **20b** faces downward.

[0071] Peak amplifier **30** is mounted on the lower surface of wiring layer **122**. Peak amplifier **30** is, for example, an amplifier including a GaN-based HEMT. Peak amplifier **30** includes a substrate and a semiconductor layer provided on or over the substrate. Peak amplifier **30** has a three dimensional shape of a rectangular parallelepiped including a main surface **30a** and a main surface **30b** opposite to main surface **30a**. Main surface **30a** is in the semiconductor layer, and main surface **30b** is in the substrate. An amplifier circuit including the semiconductor layer is formed on main surface **30a**. In main surface **30a**, a gate pad is provided in the vicinity of one of two sides parallel to each other, and a drain pad is provided in the vicinity of the other side. Further, source pads are provided on both sides of the gate pad, and the source pads are connected to electrodes provided on main surface **30b**. Main surface **30a** faces upward, and main surface **30b** faces downward.

[0072] Surface mount component **51** is, for example, a capacitor, an inductor, or a resistor element. Surface mount component **51** is mounted on the lower surface of wiring layer **122**.

[0073] Dielectric layer **131** is provided under dielectric layer **132**. Dielectric layer **131** covers the lower surfaces of wiring layer **122** and surface mount component **51**. Dielectric layer **131** covers the side surface of driver amplifier **10**, the side surface of carrier amplifier **20**, and the side surface of peak amplifier **30** from the side. Main surface **10b** of driver amplifier **10**, main surface **20b** of

carrier amplifier **20**, and main surface **30b** of peak amplifier **30** are not covered with dielectric layer **131** and are exposed from dielectric layer **131**. A thickness of dielectric layer **131** is, for example, 0.3 mm to 0.4 mm. Dielectric layer **131** has, for example, a relative dielectric constant of 3.0 to 3.7, and a dielectric loss tangent of 0.002 to 0.007.

[0074] Wiring layer **121** is provided on the lower surface of dielectric layer **131**. A thickness of wiring layer **121** is, for example, 10 μm to 120 μm . A thickness of wiring layer **121** may be 80 μm to 120 μm . Wiring layer **121** is, for example, a copper layer. As will be described in detail later, a transmission line pattern and a ground pattern are formed in wiring layer **121**.

[0075] Second layered structure **112** includes a wiring layer **123**, a dielectric layer **133**, and a wiring layer **124**. Wiring layer **123** is provided on the lower surface of dielectric layer **133**, and wiring layer **124** is provided on the upper surface of dielectric layer **133**. Dielectric layer **133** is provided between wiring layer **123** and wiring layer **124**.

[0076] Wiring layer **123** is provided on or over dielectric layer **132**. A thickness of wiring layer **123** is, for example, 10 μm to 35 μm . A thickness of wiring layer **123** may be 10 μm to 18 μm . Wiring layer **123** is, for example, a copper layer. As will be described in detail later, an electrode pattern of a capacitor and a ground pattern are formed in wiring layer **123**. The ground pattern is formed in a solid state apart from the electrode pattern. Wiring layer **123** is an example of a first wiring layer.

[0077] A thickness of dielectric layer **133** is, for example, 1 μm to 24 μm . For example, dielectric layer **133** has a relative dielectric constant of 7.0 to 10.0 and a dielectric loss tangent of 0.01 to 0.03.

[0078] A thickness of wiring layer **124** is, for example, 10 μm to 35 μm . A thickness of wiring layer **124** may be 10 μm to 18 μm . Wiring layer **124** is, for example, a copper layer. As will be described in detail later, a ground pattern is formed in a solid state in wiring layer **124**. Wiring layer **124** is an example of a second wiring layer.

[0079] Third layered structure **113** includes a dielectric layer **134**, a wiring layer **125**, a dielectric layer **135**, a wiring layer **126**, a dielectric layer **136**, and a plurality of surface mount components **52**. Wiring layer **125** is provided on the lower surface of dielectric layer **135**, and wiring layer **126** is provided on the upper surface of dielectric layer **135**. Dielectric layer **135** is provided between wiring layer **125** and wiring layer **126**.

[0080] A thickness of dielectric layer **135** is, for example, 0.1 mm to 0.2 mm. For example, dielectric layer **135** has a relative dielectric constant of 3.0 to 3.7 and a dielectric loss tangent of 0.002 to 0.007. A thickness of wiring layer **125** is, for example, 10 μm to 35 μm . The thickness of wiring layer **125** may be 20 μm to 35 μm . A thickness of wiring layer **126** is, for example, 10 μm to 45 μm . The thickness of wiring layer **126** may be 30 μm to 45 μm . Wiring layers **125** and **126** are, for example, copper layers. As will be described in detail later, a ground pattern is formed in a solid state in wiring layer **125**, and a transmission line pattern and a ground pattern are formed in wiring layer **126**.

[0081] Dielectric layer **134** is provided between dielectric layer **133** and dielectric layer **135**, covers the upper surface of wiring layer **124**, and covers the lower surface of wiring layer **125**. A thickness of dielectric layer **134** is, for example, 0.1 mm to 0.2 mm. For example, dielectric layer **134** has a relative dielectric constant of 3.0 to 3.7 and a dielectric loss tangent of 0.002 to 0.007.

[0082] Surface mount component **52** is, for example, a capacitor, an inductor, or a resistor element. Surface mount component **52** is mounted on the upper surface of wiring layer **126**.

[0083] Dielectric layer **136** is provided on or over dielectric layer **135**. Dielectric layer **136** covers the upper surfaces of wiring layer **126** and surface mount component **52**. A thickness of dielectric layer **136** is, for example, 0.4 mm to 1.0 mm. For example, dielectric layer **136** has a relative dielectric constant of 3.0 to 4.0 and a dielectric loss tangent of 0.002 to 0.012.

[0084] Conductive vias **141** and **142** for signal transmission and a conductive via **151** for grounding are provided in dielectric layer **131**. Conductive vias **141**, **142**, and **151** electrically connect a part of wiring layer **121** and a part of wiring layer **122**, respectively. The part of wiring

layer **121** to which conductive via **141** is connected is used as input terminal RFin, and the part of wiring layer **121** to which conductive via **142** is connected is used as output terminal RFout. The part of wiring layer **121** to which conductive via **151** is connected is used as a ground terminal to which a ground potential is applied.

[0085] At least one conductive via **143** for signal transmission is provided in dielectric layer **132**. Conductive via **143** electrically connects a part of wiring layer **122** and a part of wiring layer **123**.

[0086] At least one conductive via **153** for grounding is provided in dielectric layer **133**.

Conductive via **153** electrically connects a part of wiring layer **123** and a part of wiring layer **124**.

[0087] At least one conductive via **155** for grounding is provided in dielectric layer **135**.

Conductive via **155** electrically connects a part of wiring layer **125** and a part of wiring layer **126**.

[0088] A conductive via **156** for grounding is provided in dielectric layers **133**, **134**, and **135**.

Conductive via **156** electrically connects a part of wiring layer **123**, a part of wiring layer **124**, a part of wiring layer **125**, and a part of wiring layer **126**.

[0089] Dielectric layers **132**, **133**, **134**, and **135** are provided with at least one conductive via **144** for signal transmission and a conductive via **157** for grounding. Conductive via **144** electrically connects a part of wiring layer **122** and a part of wiring layer **126**. Conductive via **144** may be further electrically connected to a part of wiring layer **123**. Conductive via **157** electrically connects a part of wiring layer **122**, a part of wiring layer **123**, a part of wiring layer **124**, a part of wiring layer **125**, and a part of wiring layer **126**.

[0090] A strip line is formed by a transmission line pattern formed in wiring layer **122**, a ground pattern formed in wiring layer **123**, and a part of dielectric layer **132** therebetween. In addition, a strip line is formed by a transmission line pattern formed in wiring layer **126**, a ground pattern formed in wiring layer **125**, and a part of dielectric layer **135** therebetween. The ground pattern formed in wiring layers **123**, **124**, or **125** blocks the propagation of the electromagnetic wave generated in first layered structure **111** to third layered structure **113**, and blocks the propagation of the electromagnetic wave generated in third layered structure **113** to first layered structure **111**.

[0091] In second layered structure **112**, the electrode pattern formed in wiring layer **123** is not electrically connected to wiring layer **124**, and a thin film capacitor is formed by the electrode pattern formed in wiring layer **123**, the ground pattern formed in wiring layer **124**, and a part of dielectric layer **133** therebetween.

[0092] High frequency amplifier **1** is mounted on, for example, a printed circuit board **100** of a communication device. Printed circuit board **100** includes a base **101**, a signal wiring **102** to which input terminal RFin is connected, a signal wiring **103** to which output terminal RFout is connected, and a ground wiring **104** to which the ground terminal is connected. Signal wiring **102**, signal wiring **103**, and ground wiring **104** are provided on the upper surface of base **101**.

[0093] A part of the transmission line pattern in wiring layer **121**, which is connected to signal wiring **102** to which input terminal RFin is connected, is connected to a part (input of input matching circuit **11**) of the transmission line pattern in wiring layer **122** via conductive via **141**. Further, a part of the transmission line pattern in wiring layer **121**, which is connected to signal wiring **103** to which output terminal RFout is connected, is connected to a part (output of Doherty network **42**) of the transmission line pattern in wiring layer **122** via conductive via **142**. The ground pattern of wiring layer **121** is connected to the ground patterns in wiring layers **122**, **123**, **124**, **125**, and **126** via conductive vias **151**, **153**, **155**, **156**, and **157**, and the like. A conductive via for grounding may be provided in dielectric layer **132**.

[0094] A part of the transmission line pattern in wiring layer **122**, which is connected to the drain pad (output terminal **19**) of driver amplifier **10**, is connected to a part (input of output matching circuit **12**) of the transmission line pattern in wiring layer **126** via one conductive via **144** (a conductive via **91** in FIGS. **5** and **9**). A part of the transmission line pattern in wiring layer **126**, which is connected to the output of phase adjusting circuit **23**, is connected to a part (input of input matching circuit **21**) of the transmission line pattern in wiring layer **122** via one conductive via **144**.

(a conductive via **92** in FIGS. **5** and **9**). A part of the transmission line pattern in wiring layer **126**, which is connected to the output of phase adjusting circuit **33**, is connected to a part (input of input matching circuit **31**) of the transmission line pattern in wiring layer **122** via one conductive via **144** (a conductive via **93** in FIGS. **5** and **9**).

[0095] A part of the transmission line pattern in wiring layer **122** is connected to a part of the electrode pattern in wiring layer **123** via conductive via **143**.

[0096] Next, the layout of each wiring layer will be described. FIG. **5** is a diagram illustrating the layout of wiring layer **126**. FIG. **6** is a diagram illustrating the layout of wiring layer **125**. FIG. **7** is a diagram illustrating the layout of wiring layer **124**. FIG. **8** is a diagram illustrating the layout of wiring layer **123**. FIG. **9** is a diagram illustrating the layout of wiring layer **122**. FIG. **10** is a diagram illustrating the layout of wiring layer **121**. In FIGS. **5** to **10**, the ground pattern is provided with a matte finish (dot-like pattern). FIG. **5** also illustrates surface mount components **52**, and FIG. **9** also illustrates surface mount components **51**, driver amplifier **10**, carrier amplifier **20**, and peak amplifier **30**.

[0097] As illustrated in FIG. **5**, a ground pattern **126G**, a transmission line pattern constituting output bias circuit **14**, and a transmission line pattern constituting output matching circuit **12** are formed in wiring layer **126**. A transmission line pattern constituting branch circuit **41**, a transmission line pattern constituting phase adjusting circuit **23**, and a transmission line pattern constituting phase adjusting circuit **33** are further formed in wiring layer **126**. On the upper surface of the transmission line pattern in wiring layer **126**, capacitors **C6**, **C8**, **C23**, **C24**, **C29**, **C30**, **C32**, and **C1**, inductors **L3**, **L10**, **L4**, **L9**, **L5**, **L11**, **L12**, **L13**, **L14**, and **L15**, and resistor element **R3** are mounted as surface mount components **52**.

[0098] As illustrated in FIG. **6**, a ground pattern **125G** is formed in a solid state in wiring layer **125**. An opening through which conductive via **144** penetrates is formed in wiring layer **125**.

[0099] As illustrated in FIG. **7**, a ground pattern **124G** is formed in a solid state in wiring layer **124**. An opening through which conductive via **144** penetrates is formed in wiring layer **124**.

[0100] As illustrated in FIG. **8**, a ground pattern **123G** is formed in a solid state in wiring layer **123**. An electrode pattern of capacitor **C2** or **C4** of input matching circuit **11**, an electrode pattern of capacitor **C12** or **C7** of input matching circuit **21**, and an electrode pattern of capacitor **C5** or **C13** of input matching circuit **31** are further formed in wiring layer **123**. An opening through which conductive via **144** penetrates is formed in wiring layer **123**.

[0101] As illustrated in FIG. **9**, a ground pattern **122G**, a transmission line pattern constituting input matching circuit **11**, and a transmission line pattern constituting Doherty network **42** are formed in wiring layer **122**. A transmission line pattern constituting input matching circuit **21**, a transmission line pattern constituting input bias circuit **24**, and a transmission line pattern constituting output matching circuit **22** are further formed in wiring layer **122**. A transmission line pattern constituting input matching circuit **31**, a transmission line pattern constituting input bias circuit **34**, and a transmission line pattern constituting output matching circuit **32** are further formed in wiring layer **122**. On the lower surface of the transmission line patterns in wiring layer **122**, capacitors **C3**, **C11**, **C15**, **C9**, **C26**, **C10**, **C21**, **C28**, **C16**, and **C25**, inductors **L1**, **L2**, **L17**, **L6**, **L7**, **L8**, and **L16**, and resistor elements **R4** and **R5** are mounted as surface mount components **51**.

[0102] As illustrated in FIG. **10**, a ground pattern **121G** is formed in a solid state in wiring layer **121**.

[0103] Output terminal **19** of driver amplifier **10** and input terminal **28** of carrier amplifier **20** may be adjacent to each other. However, when the spatial distance between output terminal **19** of driver amplifier **10** and input terminal **28** of carrier amplifier **20** is small and the phase difference of the RF signal between output terminal **19** and input terminal **28** is small (including the same phase), there is a possibility that the state becomes electrically unstable. For example, oscillation may occur.

[0104] In contrast, when the phase of the RF signal is in antiphase between output terminal **19** of

driver amplifier **10** and input terminal **28** of carrier amplifier **20**, the electrically unstable state can be made less likely to occur. For example, when the electrical length between output terminal **19** of driver amplifier **10** and input terminal **28** of carrier amplifier **20** is within a range of $(2n+1)\times\pi-\pi/2$ to $(2n+1)\times\pi+\pi/2$ in terms of a phase of a wavelength λ of the RF signal input to input terminal RFin, the phase of the RF signal is close to the antiphase, and the electrically unstable state can be made less likely to occur. That is, even when driver amplifier **10** and carrier amplifier **20** are close to each other, the electrical characteristics of carrier amplifier **20** can be easily stabilized. Here, n is an integer of 0 or more.

[0105] Output terminal **19** of driver amplifier **10** and input terminal **38** of peak amplifier **30** may be adjacent to each other. However, when the spatial distance between output terminal **19** of driver amplifier **10** and input terminal **38** of peak amplifier **30** is small and the phase difference of the RF signal between output terminal **19** and input terminal **38** is small (including the same phase), there is a possibility that the state becomes electrically unstable. For example, oscillation may occur.

[0106] In contrast, when the phase of the RF signal is in antiphase between output terminal **19** of driver amplifier **10** and input terminal **38** of peak amplifier **30**, the electrically unstable state can be made less likely to occur. For example, when the electrical length between output terminal **19** of driver amplifier **10** and input terminal **38** of peak amplifier **30** is within a range of $(2n+1)\times\pi-\pi/2$ to $(2n+1)\times\pi+\pi/2$ in terms of a phase of wavelength λ of the RF signal input to input terminal RFin, the phase of the RF signal is close to the antiphase, and the electrically unstable state can be made less likely to occur. That is, even when driver amplifier **10** and peak amplifier **30** are close to each other, the electrical characteristics of peak amplifier **30** can be easily stabilized.

[0107] In the embodiment, wiring layer **126** has the transmission line pattern illustrated in FIG. 5, so that the electrical length between output terminal **19** of driver amplifier **10** and input terminal **38** of peak amplifier **30** is within a range of $(2n+1)\times\pi-\pi/2$ to $(2n+1)\times\pi+\pi/2$. Specifically, for example, in FIG. 5, the transmission line pattern from the output terminal (drain output) of driver amplifier **10** to branch circuit **41** is largely routed from the center to the right half. Further, phase adjusting circuit **23** is provided between branch circuit **41** and carrier amplifier **20**, and phase adjusting circuit **33** is provided between branch circuit **41** and peak amplifier **30**. In addition, the transmission line pattern from the output of branch circuit **41** to conductive via **92** and the transmission line pattern from the output of branch circuit **41** to conductive via **93** are not linear but curved.

[0108] Further, heat generated in driver amplifier **10** is transferred from main surface **10b** to wiring layer **121**, heat generated in carrier amplifier **20** is transferred from main surface **20b** to wiring layer **121**, and heat generated in peak amplifier **30** is transferred from main surface **30b** to wiring layer **121**. The heat transferred to wiring layer **121** is discharged to the outside via ground wiring **104** of printed circuit board **100**.

(Operation of High Frequency Amplifier)

[0109] Next, the operation of high frequency amplifier **1** will be described.

[0110] An RF signal input to input terminal RFin (a part of wiring layer **121**) via signal wiring **102** reaches wiring layer **122** via conductive via **141**, and is input to driver amplifier **10** via input matching circuit **11** provided in wiring layer **122**. The RF signal amplified by driver amplifier **10** is directed to wiring layer **126** via conductive via **91**. Then, the RF signal is transmitted to branch circuit **41** via output matching circuit **12** in wiring layer **126**, and is branched into first signal line **41A** and second signal line **41B** by branch circuit **41**.

[0111] The RF signal output to first signal line **41A** is input to phase adjusting circuit **23**. Phase adjusting circuit **23** delays the phase of the input RF signal (the input signal of carrier amplifier **20**) by a predetermined distribution constant and outputs the delayed signal. On the other hand, the RF signal output to second signal line **41B** is input to phase adjusting circuit **33**. Phase adjusting circuit **33** delays the phase of the input RF signal (the input signal of peak amplifier **30**) by a predetermined distribution constant and outputs the delayed signal. The RF signal output from

phase adjusting circuit **23** is directed to wiring layer **122** via conductive via **92** and is input to carrier amplifier **20** via input matching circuit **21**. The RF signal output from phase adjusting circuit **33** is directed to wiring layer **122** via conductive via **93** and is input to peak amplifier **30** via input matching circuit **31**.

[0112] In asymmetrical Doherty amplifier **40**, peak amplifier **30** and carrier amplifier **20** exhibit different maximum output intensities with respect to the input RF signal from each other. For example, peak amplifier **30** has a saturation output (size) that is about twice as large as that of carrier amplifier **20**, and peak amplifier **30** initiates the amplifying operation in a case where the output of carrier amplifier **20** has reached the saturation region. Specifically, carrier amplifier **20** operates in class AB or class B, and peak amplifier **30** operates in class C. When the instantaneous power is small, carrier amplifier **20** operates and peak amplifier **30** does not operate, so that the power efficiency is improved. When the instantaneous power is large, both carrier amplifier **20** and peak amplifier **30** operate, and thus the saturation power can be increased while maintaining high power efficiency. When the saturation output of peak amplifier **30** is larger than the saturation output of carrier amplifier **20**, the phase shift amount for obtaining the optimum matching in peak amplifier **30** is larger than that in carrier amplifier **20**.

[0113] For example, driver amplifier **10** has a size sufficient for 10 W output, carrier amplifier **20** has a size sufficient for 15 W output, and peak amplifier **30** has a size sufficient for 30 W output. In this case, the current consumption or the power consumption may increase in the order of driver amplifier **10**, carrier amplifier **20**, and peak amplifier **30**. The magnitude of heat generation associated with the operation may also increase in the order of driver amplifier **10**, carrier amplifier **20**, and peak amplifier **30**.

[0114] The RF signal amplified by carrier amplifier **20** is input to Doherty network **42** via output matching circuit **22**, and the RF signal amplified by peak amplifier **30** is input to Doherty network **42** via output matching circuit **32**. Then, the RF signal output from Doherty network **42** reaches output terminal RFout (a part of wiring layer **121**) via conductive via **142**, and is output via signal wiring **103**.

(Method of Manufacturing High Frequency Amplifier)

[0115] Next, a method of manufacturing high frequency amplifier **1** will be described. FIGS. **11** to **17** are cross-sectional views each illustrating the method of manufacturing the high frequency amplifier according to the embodiment.

[0116] First, as illustrated in FIG. **11**, a layered body is produced in which wiring layer **125** is formed on the lower surface of dielectric layer **135**, wiring layer **126** is formed on the upper surface of dielectric layer **135**, and conductive vias **155** are formed inside dielectric layer **135**. The layered body can be produced using, for example, a copper-clad laminate. At this time, the parts of wiring layer **126** overlapping conductive via **144**, **156**, or **157** are not formed.

[0117] As illustrated in FIG. **12**, second layered structure **112** is produced in which wiring layer **123** is formed on the lower surface of dielectric layer **133**, wiring layer **124** is formed on the upper surface of dielectric layer **133**, and conductive vias **153** are formed in dielectric layer **133**. Second layered structure **112** can be produced using, for example, a copper-clad laminate.

[0118] Next, as illustrated in FIG. **13**, dielectric layer **132**, wiring layer **122** and conductive via **143** are formed under second layered structure **112**. At this time, the parts of wiring layer **122** overlapping conductive via **144** or **157** are not formed.

[0119] Next, as illustrated in FIG. **14**, the layered body illustrated in FIG. **11** is superposed over the layered body illustrated in FIG. **13** with dielectric layer **134** being interposed therebetween.

[0120] Next, as illustrated in FIG. **15**, conductive vias **144**, **156** and **157** are formed. At this time, a part of wiring layer **126** overlapping conductive vias **144**, **156**, or **157** and a part of wiring layer **122** overlapping conductive vias **144**, **156**, or **157** are formed.

[0121] Next, as illustrated in FIG. **16**, surface mount component **52** is mounted on the upper surface of wiring layer **126**, and surface mount component **51**, driver amplifier **10**, carrier amplifier

20, and peak amplifier **30** are mounted on the lower surface of wiring layer **122**. Conductive vias **141** and **142** are provided on the lower surface of wiring layer **122**. Conductive vias **141** and **142** may be made of copper.

[0122] Next, as illustrated in FIG. **17**, dielectric layer **136** is formed on or over dielectric layer **135**, dielectric layer **131** is formed under dielectric layer **132**, and wiring layer **121** is formed under dielectric layer **131**. In forming dielectric layer **131**, a dielectric layer to be dielectric layer **131** is formed thicker than dielectric layer **131**, and then the dielectric layer is polished so that main surface **10b** of driver amplifier **10**, main surface **20b** of carrier amplifier **20**, and main surface **30b** of peak amplifier **30** are exposed. Wiring layer **121** includes a ground pattern in contact with main surface **10b** of driver amplifier **10**, main surface **20b** of carrier amplifier **20**, and main surface **30b** of peak amplifier **30**, a transmission line pattern in contact with conductive via **141**, and a transmission line pattern in contact with conductive via **142**.

[0123] In this way, high frequency amplifier **1** according to the embodiment can be manufactured.

[0124] In high frequency amplifier **1**, second layered structure **112** including wiring layer **123**, dielectric layer **133**, and wiring layer **124** is provided between driver amplifier **10**, carrier amplifier **20**, and peak amplifier **30**; and branch circuit **41**, phase adjusting circuit **23**, and phase adjusting circuit **33**. Thus, interference of signals between driver amplifier **10**, carrier amplifier **20**, and peak amplifier **30**; and branch circuit **41**, phase adjusting circuit **23**, and phase adjusting circuit **33** is less likely to occur, and the high frequency characteristics can be improved.

[0125] Further, the uniformity of the line length between capacitors **C2**, **C4**, **C7**, **C12**, **C5**, and **C13** and the input of driver amplifier **10**, carrier amplifier **20**, or peak amplifier **30** can be increased, and the high frequency characteristics can be improved.

[0126] The high frequency characteristics of the amplifiers in two examples with different forms of capacitors will be further described with reference to FIGS. **18** and **19**. FIG. **18** is a diagram illustrating a layout in a first example, and FIG. **19** is a diagram illustrating a layout in a second example.

[0127] In the first example illustrated in FIG. **18**, an amplifier **60** includes gate terminals **211**, **212**, **213** and **214**, source terminals **221** and **222**, and drain terminals **231**, **232**, **233** and **234**.

[0128] Gate terminals **211**, **212**, **213**, and **214** are arranged in this order along the first axis. Gate terminals **211**, **212**, **213** and **214** are connected to a wiring layer **61**. Wiring layer **61** is connected to a capacitor via a conductive via **71**. This capacitor is provided in second layered structure **112**, as in high frequency amplifier **1**.

[0129] Source terminals **221** and **222** are arranged in this order along the first axis. Source terminal **221** is located farther in the positive direction of the second axis, which is perpendicular to the first axis, than the row of gate terminals **211**, **212**, **213**, and **214**. Source terminal **221** is located farther in the negative direction of the first axis than gate terminal **211**, and source terminal **222** is located farther in the positive direction of the first axis than gate terminal **214**. Source terminals **221** and **222** are connected to a wiring layer **62**. Wiring layer **62** is grounded via conductive vias **721** and **722**. Conductive via **721** is located farther in the negative direction of the first axis than amplifier **60**, and conductive via **722** is located farther in the positive direction of the first axis than amplifier **60**.

[0130] Drain terminals **231**, **232**, **233**, and **234** are arranged in this order along the first axis. Drain terminals **231**, **232**, **233** and **234** are located farther in the positive direction of the second axis than the row of source terminals **221** and **222**. Drain terminal **231** is located farther in the positive direction of the first axis than source terminal **221**, and drain terminal **234** is located farther in the negative direction of the first axis than source terminal **222**. Drain terminals **231**, **232**, **233** and **234** are connected to a wiring layer **63**.

[0131] An input signal of amplifier **60** is input to wiring layer **61**. Further, the output signal of amplifier **60** is output to wiring layer **63**. In amplifier **60**, the signal flows as indicated by an arrow **70**. Driver amplifier **10**, carrier amplifier **20**, and peak amplifier **30** in high frequency amplifier **1**

may have the same structure as amplifier **60**.

[0132] In amplifier **60**, the signal line length between conductive via **71** and gate terminal **211** can be made equal to the signal line length between conductive via **71** and gate terminal **214**. Further, the signal line length between conductive via **71** and gate terminal **212** can be made equal to the signal line length between conductive via **71** and gate terminal **213**.

[0133] In the second example illustrated in FIG. **19**, a wiring layer **64** is provided, and a capacitor of a surface mount component **80** is provided between wiring layer **61** and wiring layer **64**. Wiring layer **64** is grounded via a conductive via **723**. The other configuration of the second example is the same as that of the first example.

[0134] In the second example, the difference in signal line length between the capacitor and gate terminals **211**, **212**, **213**, and **214** is larger than that in the first example.

[0135] As described above, in the first example, the difference in signal line length between each of gate terminals **211**, **212**, **213**, and **214** of amplifier **60** and the capacitor is made smaller than that in the second example, and excellent high frequency characteristics are obtained.

[0136] Also, capacitors **C2**, **C4**, **C7**, **C12**, **C5**, and **C13** are formed in the second layered structure **112**, and the surface mount components of these capacitors are not required. Thus, the number of surface mount components can be reduced as compared with the case where the surface mount components of these capacitors are used.

[0137] For example, since capacitors **C2** and **C4** are included in input matching circuit **11**, the number of surface mount components in input matching circuit **11** can be reduced. Further, since capacitors **C7** and **C12** are included in input matching circuit **21**, the number of surface mount components in input matching circuit **21** can be reduced. Further, since capacitors **C5** and **C13** are included in input matching circuit **31**, the number of surface mount components in input matching circuit **31** can be reduced.

[0138] The parasitic resistance and the parasitic inductance of capacitors **C2**, **C4**, **C7**, **C12**, **C5**, and **C13** formed in the second layered structure **112** are smaller than those in the case where a surface mount component such as a multilayer ceramic capacitor is used. Thus, matching (high frequency matching) of the RF signal can be performed with low impedance.

[0139] Furthermore, since the ground pattern of wiring layer **123** is formed in a solid state, even if the relative dielectric constant of dielectric layer **133** is high, the influence of dielectric layer **133** on the RF signal transmitted via the transmission line pattern of wiring layer **122** can be reduced. In particular, output matching circuit **22**, output matching circuit **32**, and Doherty network **42** are easily affected by parasitic capacitance caused by the surrounding dielectric layer. When a ground pattern (ground region) is provided in a part of wiring layer **123** that overlaps, in a plan view, with the transmission line pattern constituting output matching circuit **22**, a part of wiring layer **123** that overlaps, in a plan view, with the transmission line pattern constituting output matching circuit **32**, and a part of wiring layer **123** that overlaps, in a plan view, with Doherty network **42**, it is easy to reduce deterioration of the RF signal in output matching circuit **22**, output matching circuit **32**, and Doherty network **42**.

[0140] Note that, when one of phase adjusting circuits **23** and **33** is provided, the other does not need to be provided. For example, phase adjusting circuit **33** does not need to be provided as long as phase adjusting circuit **23** is provided, and phase adjusting circuit **23** does not need to be provided as long as phase adjusting circuit **33** is provided.

[0141] Further, a base member made of metal, such as copper, may be used instead of wiring layer **121**. In this case, main surface **10b** of driver amplifier **10**, main surface **20b** of carrier amplifier **20**, and main surface **30b** of peak amplifier **30** can be fixed to the base member and electrically connected to the base member by using, for example, a sintered material of silver or copper.

[0142] Although the embodiments have been described in detail, the present disclosure is not limited to the specific embodiments, and various modifications and changes can be made within the scope described in the claims.

Claims

1. A high frequency amplifier including a driver amplifier configured to amplify an input high frequency signal and an asymmetrical Doherty amplifier configured to amplify a high frequency signal output from the driver amplifier, the high frequency amplifier comprising: a first layered structure; a second layered structure layered on the first layered structure; and a third layered structure layered on the second layered structure, wherein the asymmetrical Doherty amplifier includes: a carrier amplifier; a peak amplifier configured to initiate an amplifying operation in a case where an output of the carrier amplifier has reached a saturation region, the peak amplifier having a saturation output different from a saturation output of the carrier amplifier; a branch circuit configured to receive the high frequency signal output from the driver amplifier; a first signal line provided between the branch circuit and the carrier amplifier; a second signal line provided between the branch circuit and the peak amplifier; and a phase adjusting circuit provided in either one or both of the first signal line and the second signal line and configured to delay either one or both of a phase of an input signal to the carrier amplifier and a phase of an input signal to the peak amplifier, wherein the branch circuit is configured to branch the high frequency signal output from the driver amplifier into the first signal line and the second signal line, wherein the second layered structure includes: a first wiring layer provided between the first layered structure and the third layered structure; a second wiring layer provided between the first wiring layer and the third layered structure; a dielectric layer provided between the first wiring layer and the second wiring layer; and one or more capacitors including a part of the first wiring layer, a part of the second wiring layer, and a part of the dielectric layer, wherein the driver amplifier, the carrier amplifier, and the peak amplifier are provided in the first layered structure, wherein the branch circuit and the phase adjusting circuit are provided in the third layered structure, and wherein the one or more capacitors are each electrically connected to any one of the driver amplifier, the carrier amplifier, and the peak amplifier.
2. The high frequency amplifier according to claim 1, further comprising: a first input matching circuit including the one or more capacitors and connected to an input terminal of the driver amplifier.
3. The high frequency amplifier according to claim 2, wherein the first input matching circuit is provided in the first layered structure and the second layered structure.
4. The high frequency amplifier according to claim 1, further comprising: a second input matching circuit including the one or more capacitors and connected to an input terminal of the carrier amplifier.
5. The high frequency amplifier according to claim 4, wherein the second input matching circuit is provided in the first layered structure and the second layered structure.
6. The high frequency amplifier according to claim 1, further comprising a third input matching circuit including the one or more capacitors and connected to an input terminal of the peak amplifier.
7. The high frequency amplifier according to claim 6, wherein the third input matching circuit is provided in the first layered structure and the second layered structure.
8. The high frequency amplifier according to claim 1, wherein the first wiring layer has at least a ground region overlapping, in a plan view, with a transmission line through which a high frequency signal output from the carrier amplifier or the peak amplifier is transmitted.
9. The high frequency amplifier according to claim 1, wherein when n is an integer of 0 or more, an electrical length from an output terminal of the driver amplifier to an input terminal of the carrier amplifier is within a range of $(2n+1)\times\pi-\pi/2$ to $(2n+1)\times\pi+\pi/2$, inclusive, according to a calculation in terms of a phase of the high frequency signal.
10. The high frequency amplifier according to claim 1, wherein when n is an integer of 0 or more,

an electrical length from an output terminal of the driver amplifier to an input terminal of the peak amplifier is within a range of $(2n+1)\times\pi-\pi/2$ to $(2n+1)\times\pi+\pi/2$, inclusive, in terms of a phase of the high frequency signal.

11. The high frequency amplifier according to claim 1, wherein the saturation output of the peak amplifier is larger than the saturation output of the carrier amplifier.
